

DHANUSH K

Aspiring Software Engineer | B. Tech – Information Technology

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EXECUTIVE SUMMARY

Innovative IT graduate specializing in the intersection of Embedded Systems, IoT, and Advanced AI. Hands-on experience building autonomous edge systems and cross-platform Flutter applications backed by deep learning pipelines. Deeply focused on Agentic workflows, intelligent automation, and decentralized architectures (Antigravity). Proven ability to bridge the hardware-software gap using ESP32 and Raspberry Pi. Actively seeking an entry-level role to develop next-generation, AI-driven applications and smart embedded solutions.

EDUCATION

B. Tech – information technology

Mahendra engineering college, Namakkal.

CGPA: 8.22

Year: 2022 - 2026

HSC

Govt. Boys higher secondary school, Kallakurichi.

Percentage: 81.13%

Year: 2021 - 2022

SSLC

Govt. ADW high school, Niraimathi.

Percentage: 84%

Year: 2019 - 2020

TECHNICAL SKILLS

- **Programming Languages:** Python, C, Java, Dart
- **Mobile Development:** Flutter
- **Web Technologies:** HTML, CSS, JavaScript
- **Embedded & IoT:** Raspberry Pi Zero 2W, Sensors, Wi-Fi Modules, ESP32
- **Concepts:** OOPS, Data Structures, Machine Learning, Deep Learning, Computer Vision, Agentic workflow
- **Tools:** GitHub, VS Code, Figma, Stitch, Antigravity IDE

INTERNSHIPS

PulsePath application development(Arou consultancy service)

January – October 2025

Developed an emergency ambulance routing system as a mobile application to improve response times. My Role is the mobile application was built using Flutter.

Full Stack Development Masterclass(NoviTech R&D Pvt Ltd)

April - May 2024

1. Gained one month of practical, hands-on experience in modern full-stack technologies. Developed comprehensive skills across frontend, backend, and database development.

Python + ML Foundations(TNSDC – Naan Mudhalvan | STEP NITK & Pantech E-Learning)

April 2023

1. Completed one month of foundational training in Python and Machine Learning.
2. Data Analysis & Visualization: Trained in using key data science libraries, including NumPy and Pandas for data manipulation, and Matplotlib and Seaborn for visualization.

ML & Deep Learning(Mar Baselios College of Engineering and Technology, Trivandrum)

February 2023

1. Project Achievement: Successfully built a real-time face detection tool during a 5-day intensive internship. Worked effectively as part of a 4-member team to complete the project.
2. Technology Stack: The tool was developed using Python, with OpenCV for the computer vision logic for the user interface.

PROJECTS

Smart Home Switchboard with Sound Control | ESP32, Flutter, Dart, IoT, Relays

Developed an IoT smart switchboard using an ESP32 microcontroller and a custom Flutter app, enabling appliance control via wireless commands, physical switches, and a hands-free clap sensor. Engineered a secure hardware circuit integrating a flyback SMPS power supply and optocoupler-isolated relays to safely toggle household AC appliances.

Agriculture drone monitoring system (ongoing) | IoT, ESP32 Drone Kit, CNN, wi-fi

An ai-powered drone system designed to monitor crop health, detect pests/diseases, and send real-time data for smart farming decisions.

Daily Dose'u Moble App | Dart, Flutter, Gemini API

I will build a clean, interactive, and "proper" Flutter application that helps users learn English vocabulary daily. The core feature will use AI to generate interesting, memorable stories in both English and Tamil to explain the daily word.

Smart helmet for accident detection & reporting | ESP32-CAM, SW-420, MPU6050, BLE Mobile App, Camera, wi-fi

Helmet system with impact detection, emergency alert, and real-time video logging for improved accident response and safety.

Status updater | Python, wi-fi, CNN

Created an android and desktop app to monitor customer queues and notify the shop owner when a customer approaches the billing desk.

Movie Genre Prediction Using Posters | TensorFlow, Keras, CNN

Designed and trained a convolutional neural network to classify movie genres from poster images. Implemented data preprocessing, augmentation, and multi-class classification.

ACHIEVEMENTS

- Secured Second Prize in the Movie Genre Prediction using Posters event at Idaya Engineering College.
- Won Second Prize in the Status Updater competition at Government College of Engineering, Salem.